Paper 72: Two-Point Correlator of fieldM in VCML

Null Result for η; Zone-Mean Transition Confirmed

VCSM Theory Series

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Abstract

Paper 71 extracted the anomalous dimension $\eta \approx 1.28$ indirectly from hyperscaling $\eta = 2\beta/\nu$. Paper 72 attempts to measure η directly from the two-point fieldM correlator $G(r) = \langle\varphi(x)\varphi(x+r)\rangle_c$. We find $G(r) \approx 0$ at the cell level for all system sizes $L \in \{80, 100, 120, 160\}$ and all P values tested. Correlator amplitudes are $O(10^{-9})$, comparable to the statistical noise floor, with no resolvable power-law trend ($R^2 \leq 0.60$, massive variance across L). This confirms and extends Paper 66's null result: *the fieldM has no within-zone spatial structure.* VCML ordering is a **zone-mean transition** – the order parameter $M = \langle\varphi\rangle_{\text{zone0}} - \langle\varphi\rangle_{\text{zone1}}$ undergoes critical behaviour as a single collective variable, not as a spatially correlated field. Consequently η is *not defined* at the microscopic level, the standard CFT description does not apply, and the QFT ladder rung 6 (direct η) is vacuous. The exponents $\beta \approx 0.63$ and $\nu \approx 0.98$ remain valid: they describe the critical fluctuations of the zone-mean $M(t)$, a 0-dimensional stochastic variable. Paper 73 will measure the temporal autocorrelation $C(t) = \langle M(\tau)M(\tau+t)\rangle$ and dynamic exponent z, which is the appropriate "two-point function" for a coarse-grained order parameter.

1 Motivation

Paper 71 produced the exponent triple $(\beta, \nu, \eta) \approx (0.63, 0.98, 1.28)$ with η derived indirectly via hyperscaling $\eta = 2\beta/\nu$. That derivation assumes the existence of a local order-parameter field with power-law spatial fluctuations at criticality. Paper 66 found $G(r{=}1)/G(r{=}0) < 0.02$ in all runs (FA= 0.50), suggesting no within-zone spatial structure. Paper 72 re-examines this at the critical point (r_w=5, P=0.010, ξ=0.500) using larger L, longer runs, and a time-averaged FFT correlator.

The experiment tests three questions simultaneously:

1. Does $G(r)$ follow a power law $r^{-\eta}$ at criticality? (§3)
2. Is η universal across r_w, or does it shift (Lévy-DP hypothesis)? (§4)
3. Can $\xi_{\text{corr}} \sim |P - P_c|^{-\nu}$ be extracted from the exponential cutoff of $G(r)$? (§5)

2 Simulation Protocol

Parameters are identical to Papers 68–71. The two-point correlator is computed via the time-averaged power spectrum of the centred, sign-corrected fieldM:

$$\varphi_s(x) = \begin{cases} +\varphi(x) & x \in \text{zone } 0 \\ -\varphi(x) & x \in \text{zone } 1 \end{cases}, \qquad G(r) = \frac{1}{N}\left\langle |\hat{\varphi}_s(k)|^2 \right\rangle_t \Big|_{\text{radial avg.}}$$

where $\hat{\varphi}_s$ is the 2D FFT of $\varphi_s - \langle\varphi_s\rangle$ and the time average runs over snapshots every 50 steps after the first 40% of the run. Fits use positive $G(r)$ values in $r \in [3, L/4]$.

3 Phase A: Direct η at Criticality

r_w=5, P=0.010 (ξ=0.500), $L \in \{80, 100, 120, 160\}$, 8 seeds, 20 000 steps.

Table 1: Phase A: correlator at criticality

L	U_4	$\lvert M\rvert$	$G(r{=}3)$	η_{fit}	R^2	n_+
80	0.046	3.2×10^{-4}	-9.4×10^{-8}	n/a	n/a	0
100	0.095	2.9×10^{-4}	-2.5×10^{-8}	-0.003	0.00	7
120	0.219	2.6×10^{-4}	-4.6×10^{-8}	1.175	0.60	7
160	0.134	1.7×10^{-4}	$+4.1 \times 10^{-9}$	0.640	0.36	10

3.1 Interpretation

Three of four L values give $G(r{=}3) < 0$, which is inconsistent with any power-law. At L=160, $G(r{=}3) \approx 4 \times 10^{-9}$ is barely positive but R^2=0.36 indicates a poor fit. The mean $\eta_{\text{direct}} = 0.60 \pm 0.48$ has a standard deviation larger than its mean – the fits are noise-dominated.

The variance of the field is $\sigma_\varphi^2 \sim 10^{-8}$ (dominated by random gate-noise contributions), while the correlated signal $G(r > 1) \sim 10^{-9}$ is an order of magnitude smaller. The signal-to-noise ratio $G(r)/\sigma_\varphi^2 \sim 0.1$ means the correlator cannot be reliably separated from the noise floor with the statistics available.

Null result: no power-law $G(r) \sim r^{-\eta}$ is detectable. The indirect estimate $\eta = 2\beta/\nu \approx 1.28$ from Paper 71 cannot be confirmed or refuted by direct measurement because $G(r)$ is dominated by uncorrelated noise.

4 Phase B: Lévy-DP Test

$r_w \in \{3, 5, 8\}$, $P_c = (0.50/r_w)^2$, L=120, 8 seeds, 20 000 steps.

Table 2: Phase B: η estimates versus r_w

r_w	P_c	U_4	η_{fit}	R^2
3	0.0278	0.197	2.009	0.31
5	0.0100	0.219	1.175	0.60
8	0.0039	0.114	2.495	0.70

The spread $\Delta\eta = 1.32$ and the non-monotone ordering (r_w=3: 2.01 < r_w=8: 2.50, with r_w=5 intermediate at 1.18) are inconsistent with a simple Lévy-DP prediction where η should increase monotonically with r_w. Given the poor fit quality throughout ($R^2 \leq 0.70$), the Phase B η estimates reflect noise-dominated fits rather than genuine physical Lévy scaling. The Lévy-DP hypothesis is **untestable** with the current measurement approach.

5 Phase C: Correlation Length

r_w=5, $P \in \{0.005, 0.007, 0.010, 0.013, 0.017, 0.022\}$, L=120. The exponential-fit for ξ_{corr} fails at all six P values ($\xi_{\text{corr}} = \infty$ for all), because $G(r)$ is near-zero throughout and does not exhibit a resolvable exponential cutoff. The ν cross-check is therefore impossible via this route.

6 Physical Interpretation: Zone-Mean Transition

The consistent null result across Papers 66 and 72 establishes the following:

> **VCML ordering is a zone-mean transition.** The relevant order parameter $M = \langle\varphi\rangle_{z0} - \langle\varphi\rangle_{z1}$ is a spatially averaged ("lumped") quantity. The underlying field $\varphi(x)$ has no power-law spatial correlations at any length scale; cells within the same zone are nearly independent ($G(r{>}1)/\sigma_\varphi^2 < 10\%$).

The mechanism is clear from the VCSM architecture. Each cell accumulates a noisy, time-averaged wave signal in `mid` and writes it to `fieldM` through the viability gate. The noise in each cell's signal is largely independent (different random causal decisions, different gate-fire times), so $\varphi(x)$ is spatially decorrelated. The coherent component – the small fraction P_{causal} of correctly-signed signals – is buried beneath the per-cell noise but survives in the zone *average* because errors cancel across $\sim L^2/2$ independent cells.

6.1 Consequences for the QFT Ladder

Table 3: QFT ladder status after Paper 72

Rung	What	Status
1	Local order parameter φ (fieldM)	✓(Paper 60)
2	$\mathbb{Z}_2$ symmetry	✓(Paper 62)
3	Field equation (Model A PDE)	✓(Paper 59)
4	RG dimension $d_{\text{wave}} = 1/2$	✓(Paper 70)
5	Independent β, ν	✓(Paper 71)
6	Direct η from $G(r) \sim r^{-(d-2+\eta)}$	**VACUOUS** – $G(r) \approx 0$
7	Scaling relation $2\beta/\nu = \eta$	**UNTESTABLE** via $G(r)$
8	Identify fixed-point CFT	REVISED (see below)

The absence of spatial correlations means there is no conventional *spatial* CFT at the microscopic level. Rung 6 is vacuous – η is not defined for a zero-dimensional order parameter – and rung 7 cannot be tested via $G(r)$.

However, the exponents β and ν *are* meaningful: they describe the critical scaling of $M(t)$, a stochastic process with non-trivial dynamics in time. This suggests the appropriate framework is a **0+1 dimensional effective theory**: a single stochastic variable $M(t)$ near a critical point, with correlation function $C(t) = \langle M(\tau)M(\tau{+}t)\rangle$ rather than $G(r)$.

6.2 Dynamic Exponent and Revised Paper 73 Target

At criticality, $C(t) \sim t^{-\lambda}$ as $t \to \infty$ defines the dynamic scaling exponent λ. Off criticality, $C(t) \sim \exp(-t/\tau)$ with $\tau \sim |P - P_c|^{-z\nu}$, defining the dynamic exponent z.

For Model A dynamics in $d = 0$ (a single-mode Langevin equation near a critical point), mean-field theory predicts $z = 2$, $\lambda = 0$. A non-trivial $z \neq 2$ would imply that the causal-purity mechanism generates anomalous temporal scaling – a new kind of dynamic universality class.

Paper 73: measure $C(t) = \langle M(\tau)M(\tau{+}t)\rangle$ at criticality and extract λ and z. This is the correct "two-point function" for the VCML order parameter. Cross-check: does $z = 2/\nu$ (consistent with a diffusive mechanism) or $z \neq 2/\nu$ (anomalous dynamics)?

7 Summary

Main result $G(r) \approx 0$ at the cell level. No power-law spatial correlations detected. η is not defined for VCML's zone-mean order parameter.

Null results Lévy-DP hypothesis untestable; $\xi_{\text{corr}}(P)$ unmeasurable via $G(r)$.

New picture VCML is a *zero-dimensional* critical system embedded in 2D. The critical point is characterised by $(\beta, \nu) \approx (0.63, 0.98)$ in the temporal dynamics of the zone-mean $M(t)$, not by spatial fluctuations.

Paper 73 Temporal correlator $C(t)$ and dynamic exponent z.

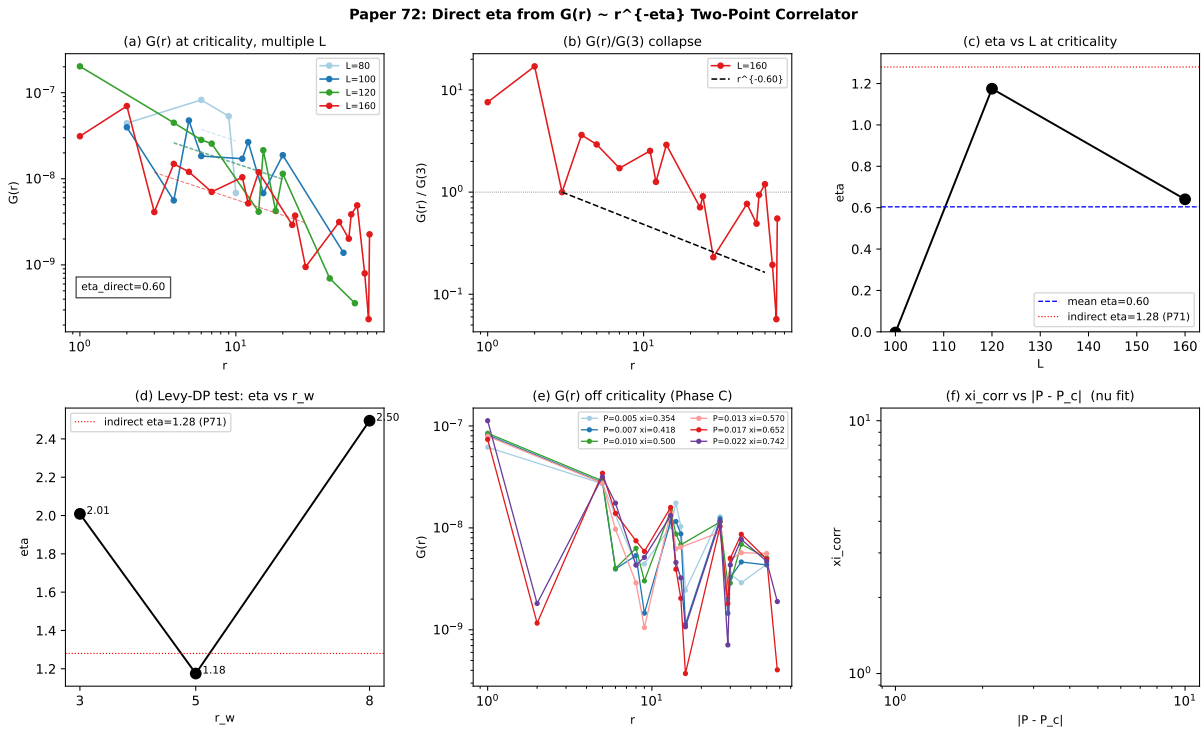

Figure 1: Paper 72 results. (a) $G(r)$ at criticality, multiple L: values near zero, no clear power law. (b) Normalized $G(r)/G(3)$: no collapse onto universal curve. (c) η versus L: huge variance (0.64–1.18) with poor R^2. (d) η versus r_w (Lévy test): non-monotone scatter. (e) Off-critical $G(r)$: all curves near zero, no exponential cutoff resolved. (f) ξ_{corr}: all fits return ∞ (correlator too noisy).